DriftCLIP

Robust, Transform-Aware Detection Using CLIP + Signals + Smart Fusion

Different Generators. Any Transform. **Real or AI?**

HACKATHON PROJECT

We build a robust detector that stays accurate under real-world transformations. Frozen CLIP features combined with carefully designed signals and fusion techniques to separate real from AI-generated images.

PROBLEM

AI images can look extremely real, but many detectors fail when images are blurred, compressed, colour - jittered or cropped. We need a detector that is robust to real-world transformations and generalises to unseen generators.

OUR APPROACH

- ✅ Frozen CLIP + small classifier (efficient & strong baseline)
- ✅ Engineering signals that capture degradation & instability
- ✅ Trained with diverse data and heavy robustness testing
- ✅ Ablations, error analysis & multi-view averaging

KEY CONTRIBUTIONS

- Rich signal suite beyond CLIP embeddings
- Robustness under 6 required transforms
- Multi-view averaging for crop tolerance
- Open-source code, reproducable & extendable

PREPROCESSING

Datasets Used

- 30,000 base data points divided into an 80:10:10 split for training, validation, and testing.
- 15 variants of post-processing transformations were applied to these base images to test robustness against real-world image degradation and manipulation for 450,000 total images.

Training	Validation	Testing
360,000 images (80%)	45,000 images (10%)	45,000 images (10%)

Data Sources

Source	Role	Images Used	Notes
WildFake	Main fake-image source	2,500 GigaGAN 2,500 DALL-E2	Balanced across GAN vs diffusion generator families
SID-SET	Social-media-realism diversity	14,000 total (12k real / 2k fake)	Real, fully-synthetic & tampered images at social-media scale
CIFAKE	Supplementary inclusion	6,000 total (3k real / 3k fake)	Native 32 x 32; kept as minority share to avoid a resolution shortcut
AIGIBench	Modern-generator diversity	2,500 StyleGAN-XL 2,500 Imagen 3	Scales detection to contemporary generator artifacts

Image Standardization Pipeline (Final Decisions)

1. Crop to square (center crop)
2. Resize to 512x512 (working resolution)
3. 1024-native fakes: small random crop (90-100% area) during training/caching only
4. Inference: crop to square → resize to 512 → feed to model (identical for all images)

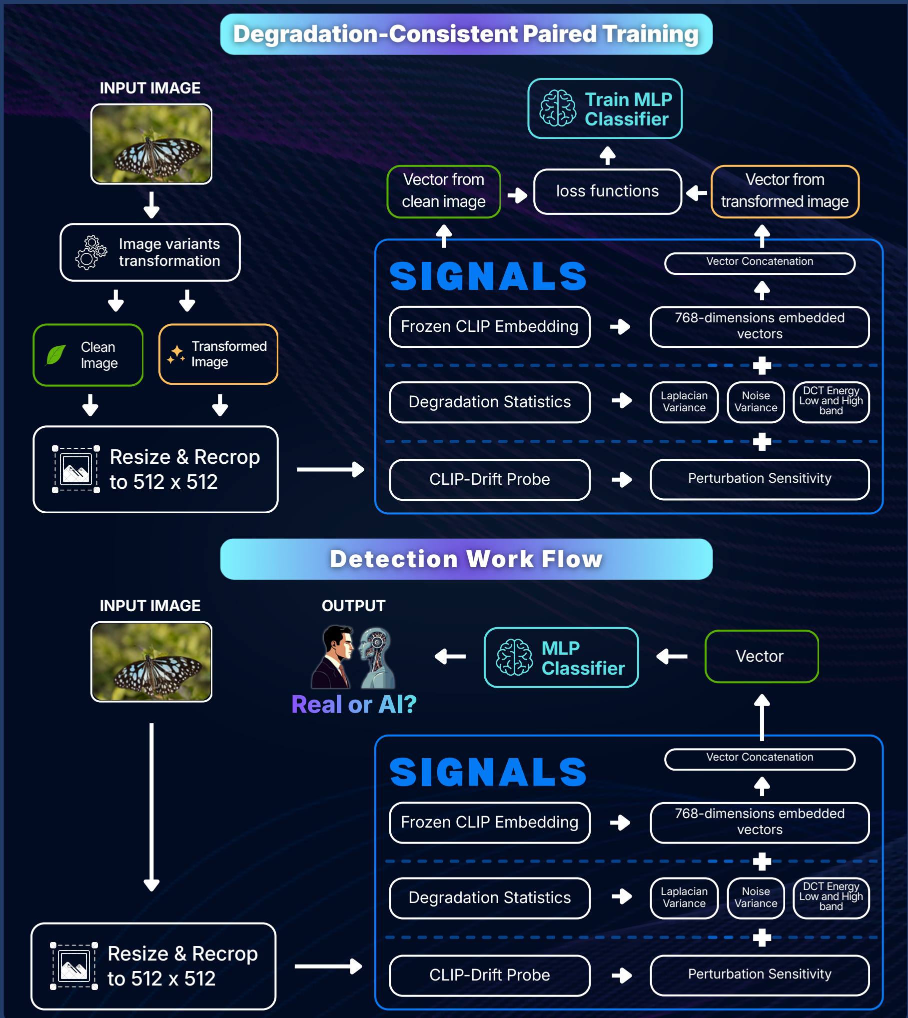
Transformed Variants (15 cached per image)

Gaussian blur	JPEG compression	Gaussian noise
Resize ⬇ then ⬆	Colour jitter	Sharpen

Sampling per training step

1 random transformed variant + 1 clean image

FLOW CHART



ROBUSTNESS SUMMARY

Model	Clean Acc. (%)	Avg. 15 Acc (%)	Acc Gap (pp)	Clean AUC	Avg. 15 AUC	AUC Gap
B32 Base	90.23	89.33	0.91	0.9563	0.9517	0.046
L14 Base	93.20	91.83	1.37	0.9650	0.9699	0.0053
L14 Concat	92.80	91.82	0.98	0.9698	0.9640	0.0058
L14 FiLM	93.27	91.55	1.71	0.9744	0.9674	0.0070
L14 Concat Drift (Final Model)	**92.90**	**91.97**	**0.93**	**0.9699**	**0.9651**	**0.0048**
L14 Concat Drift LIQ	93.13	91.79	1.34	0.9697	0.9644	0.0054

RESULT SNAPSHOT

Overall Acc.	Overall AUC	Acc Gap	AUC Gap
92.30%	0.9718	-0.013	-0.0062

MODELS

1 Embedding Model

CLIP Vit-L/14

- Vision Transformer backbone,
- Patch size 14,
- "Large" variant (~304M params)

2 Classifier Head

Simple MLP

- Chosen for non linear decision boundaries
- Fused signals (CLIP + noise / DCT stats) interact non-linearly